

- 6 (a) Fig. 6.1 shows *Vorticella*, which is a single-celled organism that lives in freshwater. *Vorticella* has many cilia which it uses for feeding.



Fig. 6.1

The distance shown by **X–Y** on Fig. 6.1 is $150\mu\text{m}$.

Calculate the magnification of Fig. 6.1.

State the formula that you will use and show your working.

Write your answer to the nearest whole number.

formula

..... [3]

- (b) The food particles are taken into the gullet by a current of water created by movement of cilia.

Any particles suspended in the water, such as bacteria, are taken into the cell as shown in Fig. 6.2.

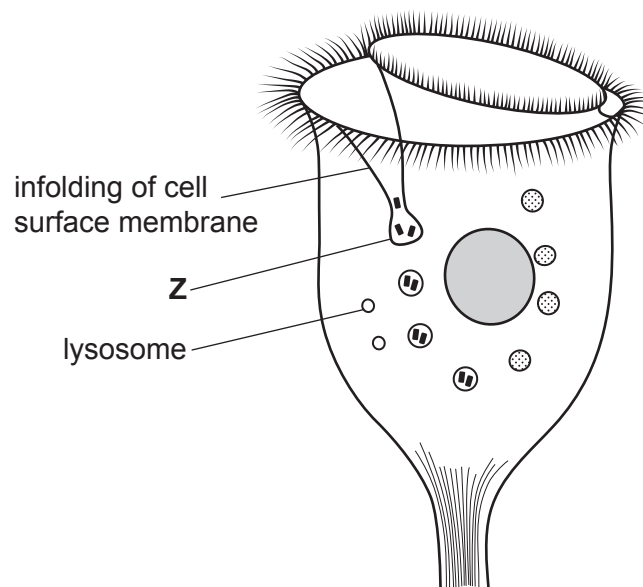


Fig. 6.2

- (i) State the name of the process which takes the bacteria into the cell at **Z** and describe the way in which it occurs.

name

description

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- (ii) Describe the role of lysosomes in intracellular digestion in *Vorticella*.

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[3]

[Total: 9]